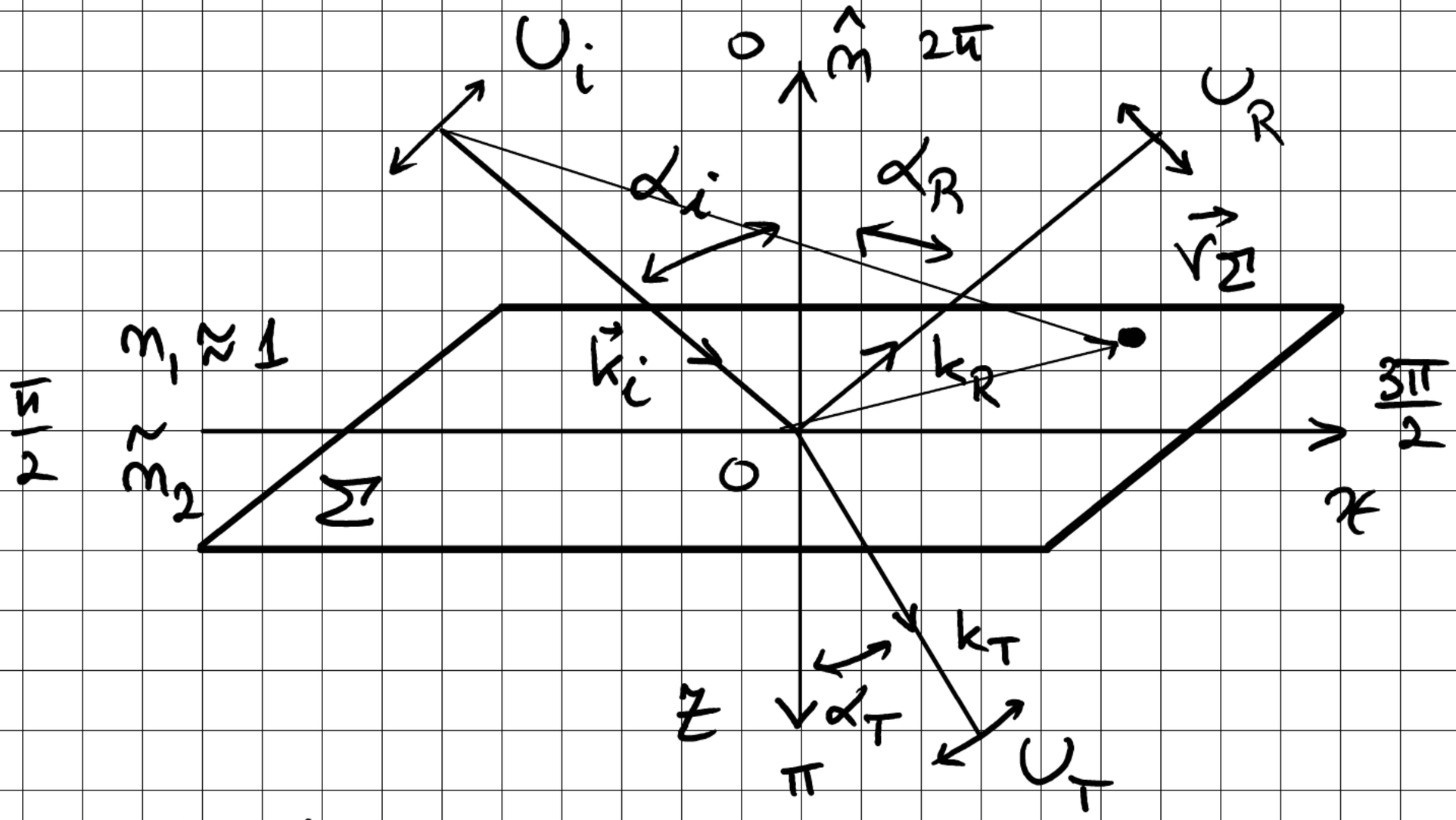


$\Sigma = (Oxy) =$ planul satelitului

$\tilde{n}_2(\lambda) = n_2(\lambda) + i k(\lambda)$

α_R

$\Rightarrow \sin(\alpha_i) = \sin(\alpha_R) \Rightarrow \alpha_i = \alpha_R \overset{not}{=} \alpha$; $\alpha_T \overset{not}{=} \beta$

α

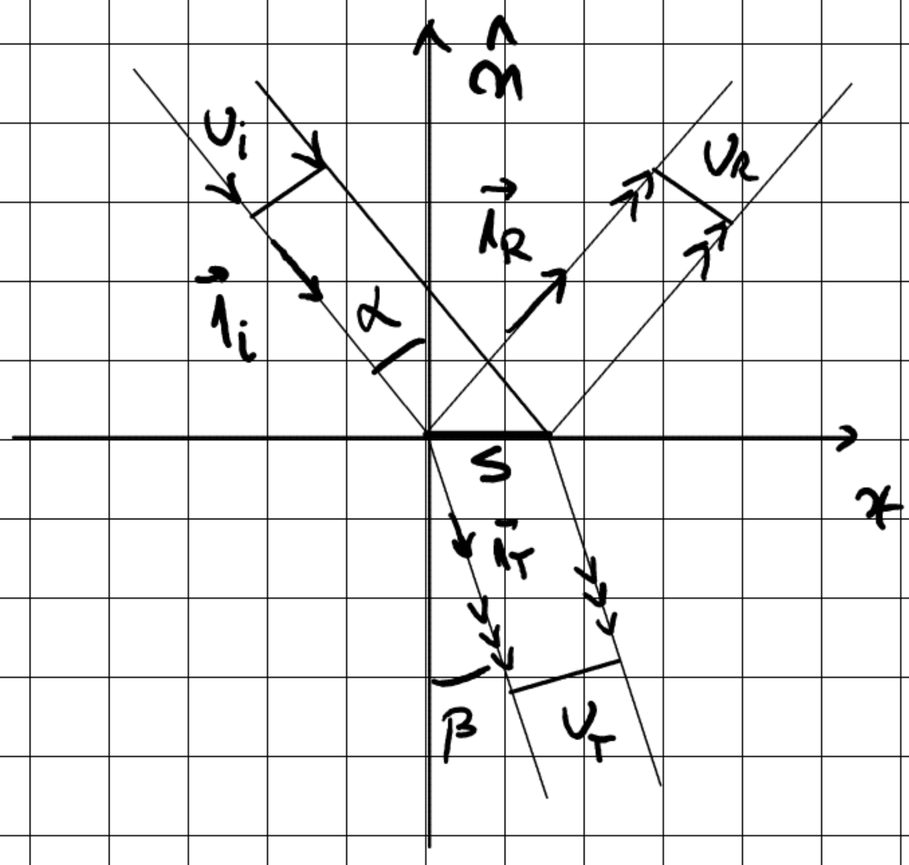
$\Rightarrow n_1 \sin(\alpha) = \tilde{n}_2(\beta)$

Snell

$$-\hat{k}_i \cdot \hat{n} = |\hat{k}_i| \cdot |\hat{n}| \cdot \cos(\alpha_i) = \cos(\alpha)$$

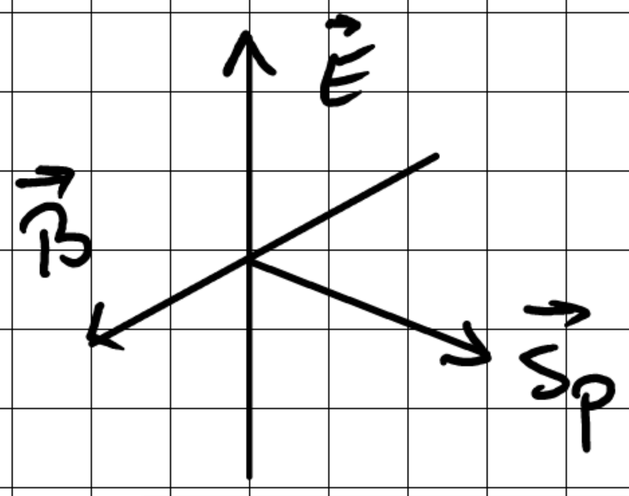
$$\boxed{\cos(\alpha) = -\hat{k}_i \cdot \hat{n}}$$

$$\cos(\alpha) \geq 0 \Leftrightarrow -\hat{k}_i \cdot \hat{n} \geq 0 \Rightarrow \underline{\text{Suprafață iluminată}}$$



$$\Phi = \oiint_S \langle \vec{S}_p \rangle d\vec{S} = \oiint_S \langle \vec{S}_p \rangle \hat{n}\, dS$$

$$= |\langle \vec{S}_p \rangle| \cdot |\hat{n}| \cos(\vartheta) \oiint_S dS$$



$$\vec{S_p} = \frac{1}{\mu_0}(\vec{E}\times\vec{B})\ \left[\frac{W}{m^2}\right]$$

$$I = \langle|\vec{S_p}|\rangle$$

$$\Rightarrow \boxed{\Phi = I\cdot\cos(\theta)S}\ ;\ \theta = \sphericalangle(\vec{S_p};\hat{n})$$

$$q = \frac{\Phi_R}{\Phi_i} = \frac{I_R\cancel{\cos(\alpha_R)}S}{I_i\cancel{\cos(\alpha_i)}S} = \frac{I_R}{I_i} \Rightarrow \boxed{q = \frac{I_R}{I_i}}$$

$$I \propto E^2$$

$$\begin{cases} U_i: \vec{E_i} = \vec{A}e^{i(\vec{k_i}\cdot\vec{r_\Sigma} - \omega_i t)} \\ U_R: \vec{E_R} = \vec{R}e^{i(\vec{k_R}\cdot\vec{r_\Sigma} - \omega_R t)} \\ U_T: \vec{E_T} = \vec{T}e^{i(\vec{k_T}\cdot\vec{r_\Sigma} - \omega_T t)} \end{cases} \quad \omega_i = \omega_R = \omega_T = \omega \Rightarrow$$

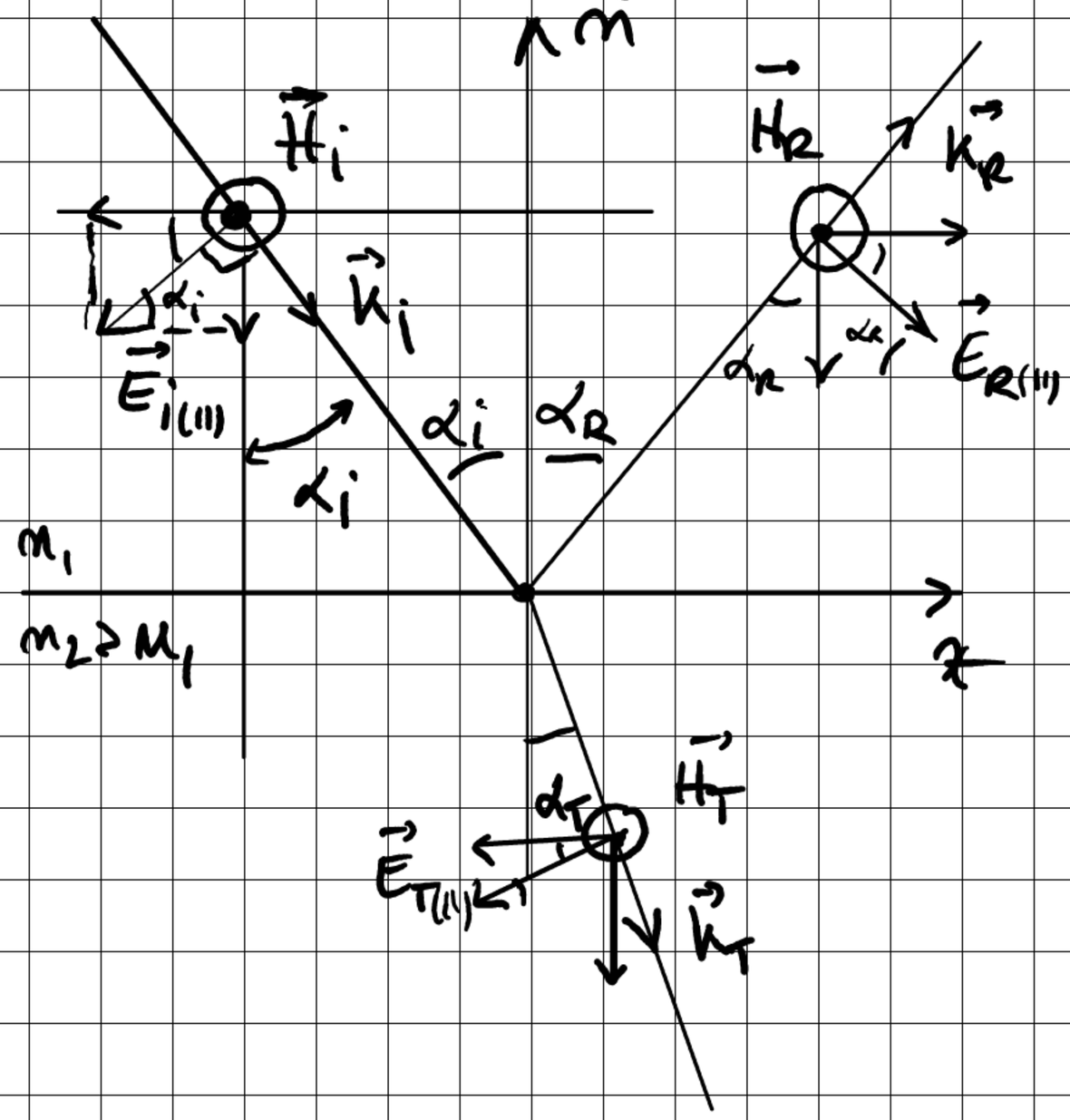
$$\Rightarrow \begin{cases} I_R \propto E_R^2 = R^2 = R_\parallel^2 + R_\perp^2 \\ I_i \propto E_i^2 = A^2 = A_\parallel^2 + A_\perp^2 \end{cases} \Rightarrow q = \frac{R_\parallel^2 + R_\perp^2}{A_\parallel^2 + A_\perp^2}$$

$$\boxed{q_\parallel = \frac{I_{R(\parallel)}}{I_{i(\parallel)}} = \frac{R_\parallel^2}{A_\parallel^2}\ ;\ q_\perp = \frac{I_{R(\perp)}}{I_{i(\perp)}} = \frac{R_\perp^2}{A_\perp^2}}$$

Formulele Fresnel

Relațiile dau posibilitatea calculului intensităților incidente și reflectate.

Ye obtin impunând cond. de cont. la nivelul Σ.



$$\begin{cases} \vec{E}_{i\parallel} = \vec{A}_\parallel e^{i(\vec{k}_1 \cdot \vec{r}_\Sigma - \omega t)} \\ \vec{E}_{i\perp} = \vec{A}_\perp e^{i(\vec{k}_1 \cdot \vec{r}_\Sigma - \omega t)} \end{cases} \quad \begin{cases} \vec{E}_{R\parallel} = \vec{R}_\parallel e^{i(\vec{k}_1 \cdot \vec{r}_\Sigma - \omega t)} \\ \vec{E}_{R\perp} = \vec{R}_\perp e^{i(\vec{k}_1 \cdot \vec{r}_\Sigma - \omega t)} \end{cases}$$

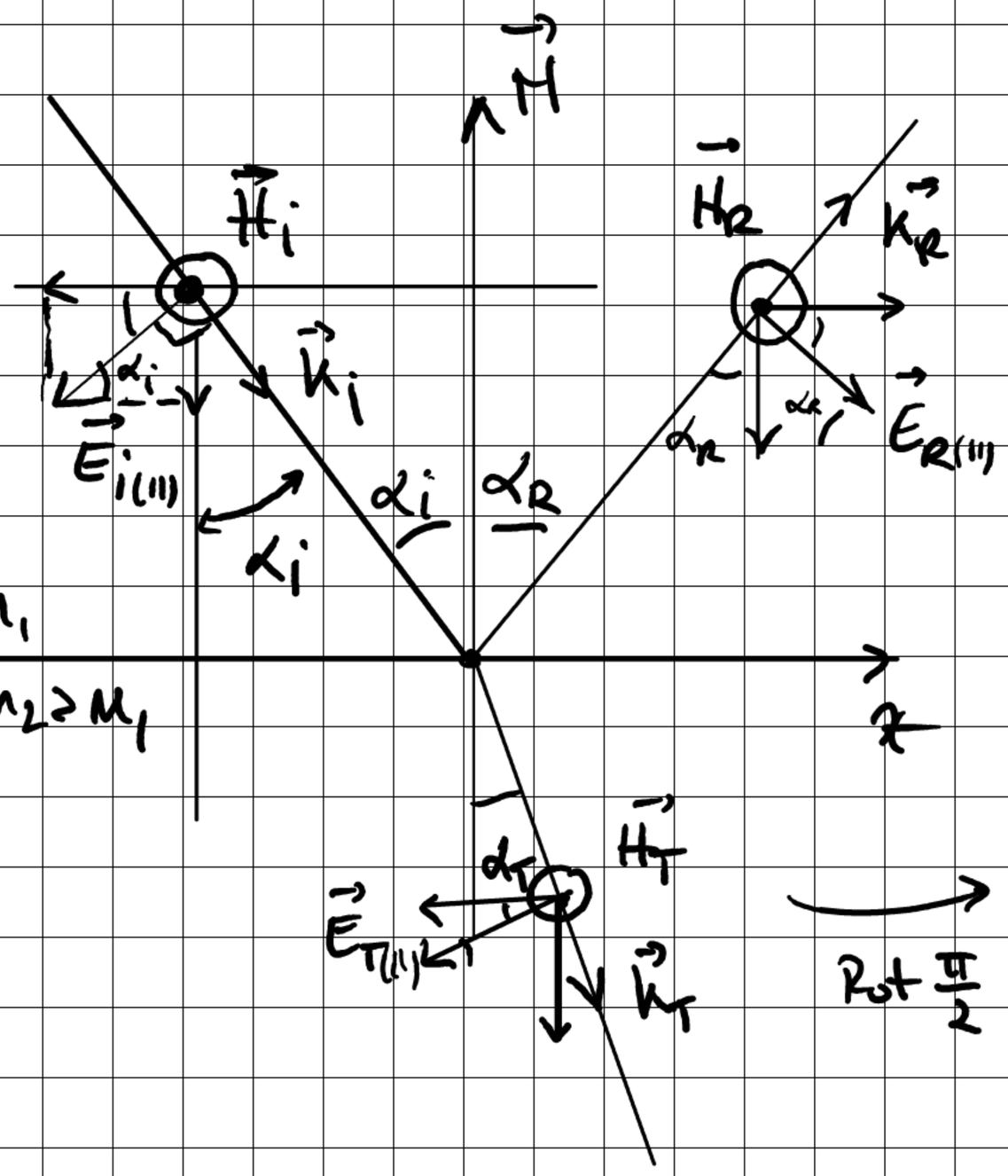
$$\begin{cases} \vec{E}_{T\parallel} = \vec{T}_\parallel e^{i(\vec{k}_2 \cdot \vec{r}_\Sigma - \omega t)} \\ \vec{E}_{T\perp} = \vec{T}_\perp e^{i(\vec{k}_2 \cdot \vec{r}_\Sigma - \omega t)} \end{cases}$$

$$\left.\begin{array}{l} \vec{H}_{i\parallel} + \vec{H}_{R\parallel} = \vec{H}_{T\parallel} \\ H = E \cdot \sqrt{\dfrac{\varepsilon}{\mu}} \end{array}\right\} \Rightarrow E_{i(\parallel)}\sqrt{\frac{\varepsilon_1}{\mu_0}} + E_{R(\parallel)}\sqrt{\frac{\varepsilon_1}{\mu_0}} = E_{T(\parallel)}\sqrt{\frac{\varepsilon_2}{\mu_0}}$$

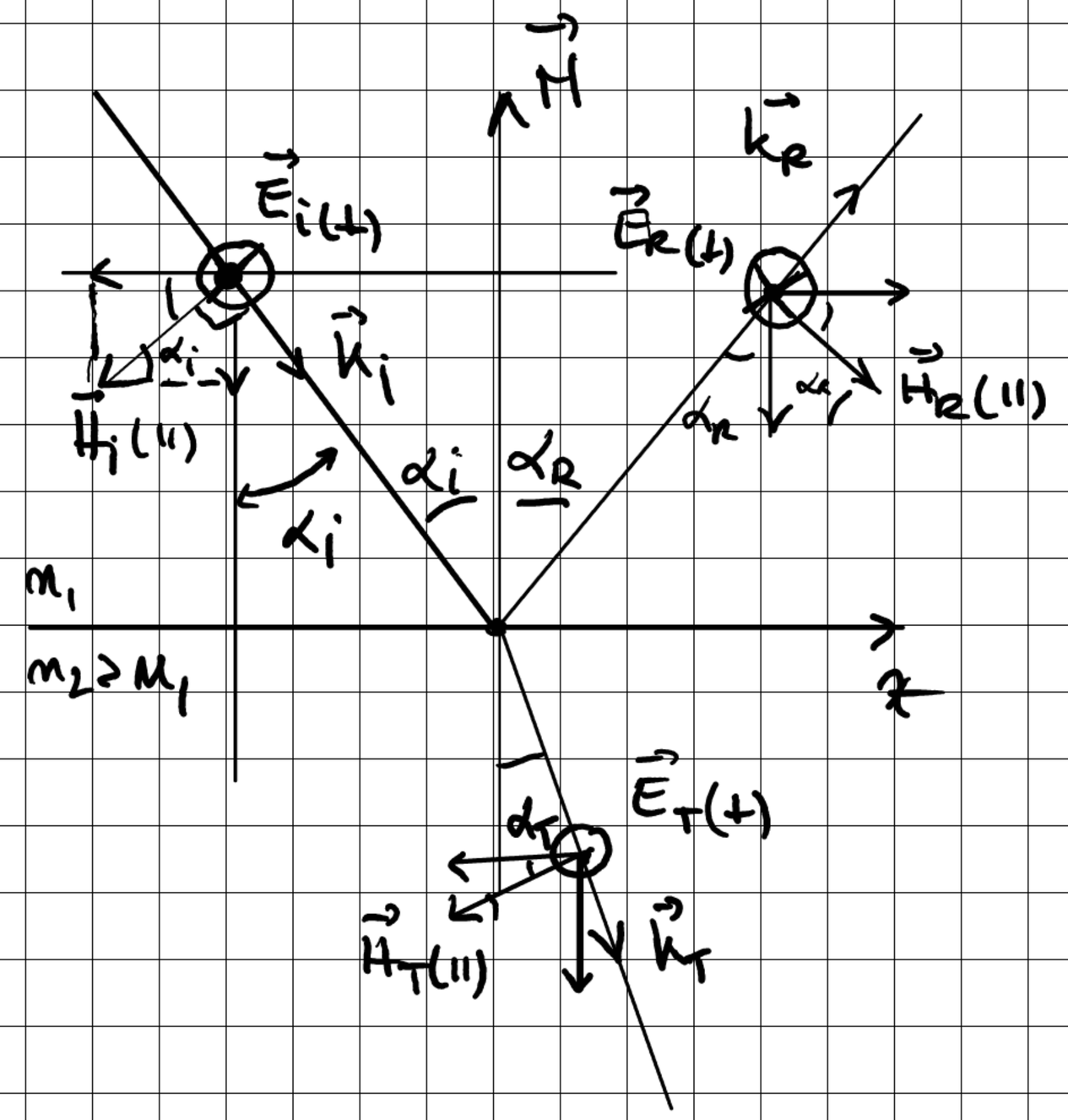
$$E_{i(\parallel)}\sqrt{\varepsilon_1} + E_{R(\parallel)}\sqrt{\varepsilon_1} = E_{T(\parallel)}\sqrt{\varepsilon_2}$$

$$\sqrt{\varepsilon_1}\left(E_{i(\parallel)} + E_{R(\parallel)}\right) = E_{T(\parallel)}\sqrt{\varepsilon_2}$$

$$-E_{i(\parallel)}\cos(\alpha_i) + E_{R(\parallel)}\cos(\alpha_R) = -E_T\cos(\alpha_T)$$



$R_0 + \frac{\pi}{2}$



$$-H_{i(\parallel)}\cos(\alpha_i) + H_{R(\parallel)}\cos(\alpha_R) = -H_{T(\parallel)}\cos(\alpha_T)$$

$$-E_{i(\perp)}\cos(\alpha_i)\sqrt{\frac{\varepsilon_1}{\mu_0}} + E_{R(\perp)}\cos(\alpha_R)\sqrt{\frac{\varepsilon_1}{\mu_0}} = -E_{T(\perp)}\cos(\alpha_T)\sqrt{\frac{\varepsilon_2}{\mu_0}}$$

$$\begin{cases} \sqrt{\varepsilon_1}\,E_{i(\parallel)} + \sqrt{\varepsilon_1}\,E_{R(\parallel)} = \sqrt{\varepsilon_2}\,E_{T(\parallel)} \\ E_{i(\perp)} + E_{R(\perp)} = E_{T(\perp)} \\ -E_{i(\parallel)}\cos(\alpha) + E_{R(\parallel)}\cos(\alpha) = -E_{T(\parallel)}\cos(\beta) \\ -E_{i(\perp)}\cos(\alpha)\sqrt{\varepsilon_1} + E_{R(\perp)}\cos(\alpha)\sqrt{\varepsilon_1} = -E_{T(\perp)}\cos(\beta)\sqrt{\varepsilon_2} \end{cases}$$

$$\sqrt{\varepsilon_1}A_\parallel e^{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)} + \sqrt{\varepsilon_1}R_\parallel e^{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)} = \sqrt{\varepsilon_2}T_\parallel e^{i(\vec{k}_2\cdot\vec{r}_\Sigma - \omega t)}$$

$$\sqrt{\varepsilon_1}e^{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)}(A_\parallel + R_\parallel) = \sqrt{\varepsilon_2}T_\parallel e^{i(\vec{k}_2\cdot\vec{r}_\Sigma - \omega t)}$$

$$\sqrt{\varepsilon_1}e^{i\vec{k}_1\vec{r}_\Sigma}\cdot\cancel{e^{-i\omega t}}(A_\parallel + R_\parallel) = \sqrt{\varepsilon_2}T_\parallel e^{i\vec{k}_2\cdot\vec{r}_\Sigma}\cdot\cancel{e^{-i\omega t}}$$

$$\sqrt{\varepsilon_1}e^{i\vec{k}_1\cdot\vec{r}_\Sigma}(A_\parallel + R_\parallel) = \sqrt{\varepsilon_2}T_\parallel e^{i\vec{k}_2\cdot\vec{r}_\Sigma}$$

$$E_{i(11)}\cos(\alpha) - E_{R(11)}\cos(\alpha) = E_{T(11)}\cos(\beta)$$

$$\cos(\alpha)\left(E_{i(11)} - E_{R(11)}\right) = E_{T(11)}\cos(\beta)$$

$$\cos(\alpha)\left(A_{11}e^{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)} - R_{11}e^{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)}\right) = T_{11}e^{i(\vec{k}_2\cdot\vec{r}_\Sigma - \omega t)}\cdot\cos(\beta)$$

$$\begin{cases}\cos(\alpha)e^{i\vec{k}_1\cdot\vec{r}_\Sigma}\left(A_{11} - R_{11}\right) = T_{11}e^{i\vec{k}_2\cdot\vec{r}_\Sigma}\cos(\beta)\\ \sqrt{\varepsilon_1}\,e^{i\vec{k}_1\cdot\vec{r}_\Sigma}\left(A_{11} + R_{11}\right) = \sqrt{\varepsilon_2}\,T_{11}\,e^{i\vec{k}_2\cdot\vec{r}_\Sigma}\end{cases}$$

$$\begin{cases}\cos(\alpha)\left(A_{11} - R_{11}\right) = T_{11}\cos(\beta)\\ \sqrt{\varepsilon_1}\left(A_{11} + R_{11}\right) = \sqrt{\varepsilon_2}\,T_{11}\end{cases} \Rightarrow \begin{cases}A_{11} - R_{11} = T_{11}\dfrac{\cos(\beta)}{\cos(\alpha)}\\ A_{11} + R_{11} = T_{11}\sqrt{\dfrac{\varepsilon_2}{\varepsilon_1}} = T_{11}\dfrac{\sin(\alpha)}{\sin(\beta)}\end{cases}$$

$$2A_{11} = T_{11}\left(\frac{\cos(\beta)}{\cos(\alpha)} + \frac{\sin(\alpha)}{\sin(\beta)}\right)$$

$$2A_{11} = T_{11}\cdot\frac{\cos(\beta)\sin(\beta) + \sin(\alpha)\cos(\alpha)}{\sin(\beta)\cos(\alpha)} \Rightarrow T_{11} = A_{11}\frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)\cos(\alpha-\beta)}$$

j.e.d

$$\frac{A_{11} - R_{11}}{A_{11} + R_{11}} = \frac{\cos(\beta)\sin(\beta)}{\cos(\alpha)\sin(\alpha)}$$

$$A_{11}\cos(\alpha)\sin(\alpha) - R_{11}\cos(\alpha)\sin(\alpha) = \left(A_{11} + R_{11}\right)\cos(\beta)\sin(\beta)$$

$$A_{11}\left(\cos(\alpha)\sin(\alpha) - \cos(\beta)\sin(\beta)\right) = R_{11}\left(\cos(\beta)\sin(\beta) + \cos(\alpha)\sin(\alpha)\right)$$

$$R_{11} = \frac{\cos(\alpha)\sin(\alpha) - \cos(\beta)\sin(\beta)}{\cos(\beta)\sin(\beta) + \cos(\alpha)\sin(\alpha)}A_{11} =$$

$$= \frac{\sin(2\alpha) - \sin(2\beta)}{\sin(2\alpha) + \sin(2\beta)}A_{11} =$$

$$= \frac{\cancel{2}\sin\left(\frac{2\alpha-2\beta}{\cancel{2}}\right)\cos\left(\frac{2\alpha+2\beta}{\cancel{2}}\right)}{\cancel{2}\sin\left(\frac{2\alpha+2\beta}{\cancel{2}}\right)\cos\left(\frac{2\alpha-2\beta}{\cancel{2}}\right)} A_{11} =$$

$$= \frac{\sin(\alpha-\beta)\cos(\alpha+\beta)}{\sin(\alpha+\beta)\cos(\alpha-\beta)} A_{11} = tg(\alpha-\beta)\,ctg(\alpha+\beta)\,A_{11}$$

$$= \frac{tg(\alpha-\beta)}{tg(\alpha+\beta)} A_{11} \quad q.e.d$$

$$A_{(1)} e^{\cancel{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)}} + R_{(1)} e^{\cancel{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)}} = T_{(1)} e^{\cancel{i(\vec{k}_2\cdot\vec{r}_\Sigma - \omega t)}}$$

$$A_{(1)} + R_{(1)} = T_{(1)}$$

$$-E_{i(1)}\cos(\alpha)\sqrt{\varepsilon_1} + E_{R(1)}\cos(\alpha)\sqrt{\varepsilon_1} = -E_{T(1)}\cos(\beta)\sqrt{\varepsilon_2} \;\big|\cdot(-1)$$

$$\sqrt{\varepsilon_1}\cos(\alpha)\left(E_{i(1)} - E_{R(1)}\right) = \sqrt{\varepsilon_2}\cos(\beta)\,E_{T(1)}$$

$$\sqrt{\varepsilon_1}\cos(\alpha)\,e^{\cancel{i(\vec{k}_1\cdot\vec{r}_\Sigma - \omega t)}}\left(A_\perp - R_\perp\right) = \sqrt{\varepsilon_2}\cos(\beta)\,T_1\,e^{\cancel{i(\vec{k}_2\cdot\vec{r}_\Sigma - \omega t)}}$$

$$\sqrt{\varepsilon_1}\cos(\alpha)\left(A_\perp - R_\perp\right) = \sqrt{\varepsilon_2}\cos(\beta)\,T_\perp$$

$$\begin{cases} A_\perp - R_\perp = \dfrac{\cos(\beta)}{\cos(\alpha)}\sqrt{\dfrac{\varepsilon_2}{\varepsilon_1}}\,T_\perp \\ A_\perp + R_\perp = T_\perp \end{cases}$$

$$2A_\perp = T_\perp\left(1 + \frac{\cos(\beta)\sin(\alpha)}{\cos(\alpha)\sin(\beta)}\right)$$

$$2A_\perp = T_\perp \cdot \frac{\cos(\alpha)\sin(\beta) + \cos(\beta)\sin(\alpha)}{\cos(\alpha)\sin(\beta)}$$

$$T_\perp = \frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)} \quad \text{q.e.d}$$

$$\frac{A_\perp - R_\perp}{A_\perp + R_\perp} = \frac{\cos(\beta)}{\cos(\alpha)}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = \frac{\cos(\beta)\sin(\alpha)}{\cos(\alpha)\sin(\beta)}$$

$$(A_\perp - R_\perp)\cos(\alpha)\sin(\beta) = (A_\perp + R_\perp)\cos(\beta)\sin(\alpha)$$

$$A_\perp(\cos(\alpha)\sin(\beta) - \cos(\beta)\sin(\alpha)) = R_\perp(\cos(\beta)\sin(\alpha) + \cos(\alpha)\cdot \sin(\beta))$$

$$R_\perp = A_\perp \cdot \frac{\cos(\alpha)\sin(\beta) - \cos(\beta)\sin(\alpha)}{\cos(\beta)\sin(\alpha) + \cos(\alpha)\sin(\beta)}$$

$$R_\perp = A_\perp \frac{\sin(\beta-\alpha)}{\sin(\alpha+\beta)} = -\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)} A_\perp \quad \text{q.e.d}$$

$$T_\perp = \frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)} A_\perp$$

$$R_\perp = -\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)} A_\perp$$

Relatiile lui Fresnel.

$$T_\parallel = \frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)\cos(\alpha-\beta)} A_\parallel$$

$$R_\parallel = \frac{tg(\alpha-\beta)}{tg(\alpha+\beta)} A_\parallel$$

$$r_\perp = \frac{R_\perp}{A_\perp} = -\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)}$$

$$t_\perp = \frac{T_\perp}{A_\perp} = \frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)}$$

$$r_{\parallel} = \frac{R_\parallel}{A_{\parallel}} = \frac{tg(\alpha-\beta)}{tg(\alpha+\beta)}$$

$$t_{\parallel} = \frac{T_\parallel}{A_{\parallel}} = \frac{2\sin(\beta)\cos(\alpha)}{\sin(\alpha+\beta)\cos(\alpha-\beta)}$$

Coef. lui Fresnel

$$q_{\parallel} = \frac{I_{R(\parallel)}}{I_{i(\parallel)}} = \frac{R_\parallel^2}{A_\parallel^2} \quad ; \quad q_\perp = \frac{I_{R(\perp)}}{I_{i(\perp)}} = \frac{R_\perp^2}{A_\perp^2}$$

$$\Rightarrow$$

$$q_{\parallel} = |r_{\parallel}|^2 = \left(\frac{tg(\alpha-\beta)}{tg(\alpha+\beta)}\right)^2 \quad ; \quad q_\perp = |r_\perp|^2 = \left(\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)}\right)^2$$

$$n_1 \sin(\alpha) = \tilde{n}_2 \sin(\beta) \Rightarrow n_1 = \tilde{n}_2 \frac{\sin(\beta)}{\sin(\alpha)}$$

$$\Rightarrow \begin{cases} r_\perp = \dfrac{\cancel{n_1}^{1}\cos(\alpha) - \tilde{n}_2\cos(\beta)}{\cancel{n_1}_{1}\cos(\alpha) + \tilde{n}_2\cos(\beta)} \\ r_{\parallel} = \dfrac{\tilde{n}_2\cos(\alpha) - \cancel{n_1}^{1}\cos(\beta)}{\tilde{n}_2\cos(\alpha) + \cancel{n_1}_{1}\cos(\beta)} \end{cases}$$

$$q_{\parallel} = \frac{R_{\parallel}^2}{A_{\parallel}^2} \; ; \; q_{\perp} = \frac{R_{\perp}^2}{A_{\perp}^2} \Rightarrow \begin{cases} R_{\parallel}^2 = q_{\parallel} A_{\parallel}^2 \\ R_{\perp}^2 = q_{\perp} A_{\perp}^2 \end{cases}$$

$$q = \frac{R_{\parallel}^2 + R_{\perp}^2}{A_{\parallel}^2 + A_{\perp}^2} = \frac{q_{\parallel} A_{\parallel}^2 + q_{\perp} A_{\perp}^2}{A_{\parallel}^2 + A_{\perp}^2}$$

Consideram radiația incidentă nepolarizată $\Rightarrow$

$$\Rightarrow A_{\parallel}^2 = A_{\perp}^2 = A^2$$

$$\boxed{q = \frac{q_{\parallel} + q_{\perp}}{2}}$$

$$\begin{cases} q_{\perp} = \left| \dfrac{\cos(\alpha) - \tilde{n}_2 \cos(\beta)}{\cos(\alpha) + \tilde{n}_2 \cos(\beta)} \right|^2 \\ q_{\parallel} = \left| \dfrac{\tilde{n}_2 \cos(\alpha) - \cos(\beta)}{\tilde{n}_2 \cos(\alpha) + \cos(\beta)} \right|^2 \end{cases}$$

$$q(\alpha; \lambda) = \frac{1}{2} \left(\left| \frac{\tilde{n}_2 \cos(\alpha) - \cos(\beta)}{\tilde{n}_2 \cos(\alpha) + \cos(\beta)} \right|^2 + \left| \frac{\cos(\alpha) - \tilde{n}_2 \cos(\beta)}{\cos(\alpha) + \tilde{n}_2 \cos(\beta)} \right|^2 \right)$$

$$\tilde{n}_2(\lambda) = n(\lambda) + i k(\lambda)$$

$$\sin(\alpha) = \tilde{n}_2 \sin(\beta) \Rightarrow \sin(\beta) = \frac{1}{\tilde{n}_2} \sin(\alpha)$$

$$\sin^2(\beta) + \cos^2(\beta) = 1 \Rightarrow \cos^2(\beta) = 1 - \sin^2(\beta) \Rightarrow$$

$$\Rightarrow \cos(\beta) = \sqrt{1 - \sin^2(\beta)}$$

$$\boxed{\cos(\beta) = \left[1 - \left(\frac{1}{\tilde{n}_2} \sin(\alpha) \right)^2 \right]^{1/2}}$$

Soarele nu este monocromatic

$I_\odot$ = iradiația spectrală solară = $I_\odot(\lambda)\left[\frac{W}{m^2 \cdot mm}\right]$

$$I_{\odot,total} = \int_{\lambda_{min}}^{\lambda'_{max}} I_\odot(\lambda)\,d\lambda$$

$$\Phi_\odot = I_{\odot,total}\cos(\alpha)S = \int_{\lambda_{min}}^{\lambda_{max}} I_\odot(\lambda)\cos(\alpha)S\,d\lambda =$$

$$= \cos(\alpha)S\int_{\lambda_{min}}^{\lambda_{max}} I_\odot(\lambda)\,d\lambda = \Phi_\odot^i$$

$$q_\odot(\alpha) = \frac{\Phi_\odot^R}{\Phi_\odot^i} \Rightarrow \Phi_\odot^R = q(\alpha,\lambda)\cdot\Phi_\odot^i$$

$$d\Phi_\odot^R = q(\alpha,\lambda)\,d\Phi_\odot^i = q(\alpha,\lambda)\cos(\alpha)S\,I_\odot(\lambda)\,d\lambda$$

$$\Phi_\odot^R = \int_{\lambda_{min}}^{\lambda_{max}} q(\alpha,\lambda)\cos(\alpha)S\,I_\odot(\lambda)\,d\lambda$$

$$= \cos(\alpha)S\int_{\lambda_{min}}^{\lambda_{max}} q(\alpha,\lambda)\,I_\odot(\lambda)\,d\lambda$$

$$q_\odot(\alpha) = \frac{\cancel{\cos(\alpha)}S\int q(\alpha,\lambda)\,I_\odot(\lambda)\,d\lambda}{\cancel{\cos(\alpha)}S\int I_\odot(\lambda)\,d\lambda}\;;\; \alpha\in[\lambda_{min};\lambda_{max}]$$

$$\rho_\odot(\alpha) = \frac{\int \rho(\alpha,\lambda)\, I_\odot(\lambda)\, d\lambda}{\int I_\odot(\lambda)\, d\lambda} \; ; \; \alpha \in [\lambda_{min}; \lambda_{max}]$$

↓ reflectanța efectivă solară

$$\rho(\alpha;\lambda) = \frac{1}{2}\left(\left| \frac{\tilde{m}_2 \cos(\alpha) - \cos(\beta)}{\tilde{m}_2 \cos(\alpha) + \cos(\beta)} \right|^2 + \left| \frac{\cos(\alpha) - \tilde{m}_2 \cos(\beta)}{\cos(\alpha) + \tilde{m}_2 \cos(\beta)} \right|^2 \right)$$

$$\cos(\beta) = \left[1 - \left(\frac{1}{\tilde{m}_2} \sin(\alpha) \right)^2 \right]^{1/2}$$

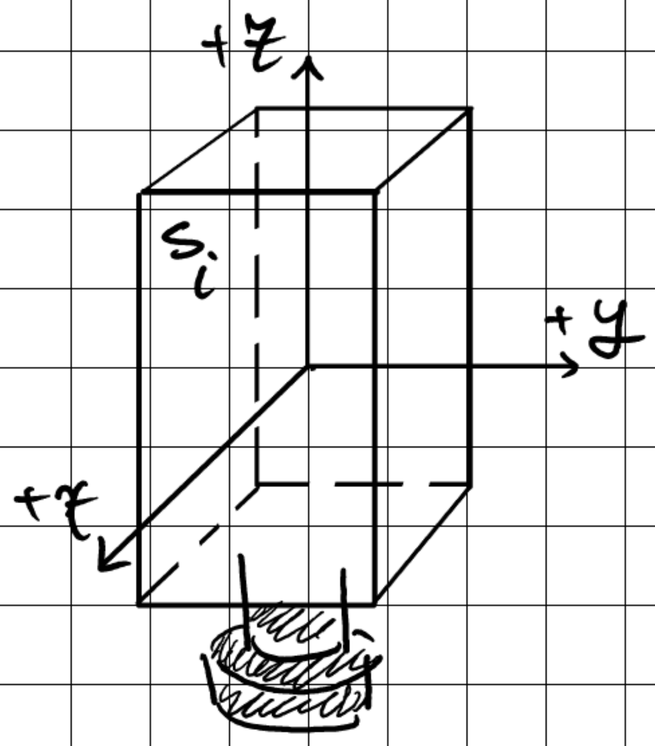
mat. $+\lambda+\alpha \rightarrow \rho(\lambda,\alpha)$

$\rho(\lambda;\alpha) + I_\odot(\lambda) \rightarrow \rho_\odot(\alpha)$

$$\tilde{m}_2(\lambda) = n_2(\lambda) + i\,k(\lambda)$$

Modelul optic pentru o suprafață ?

Exhindem modelul optic pentru N suprafete.



$$\mathcal{Y}_j \overset{not}{=} \{ S_j ; \hat{n}_j ; \varrho_{\odot,j} \} ; \; j \in \overline{1,N}$$

$$\tilde{n}_2(\lambda)_j = n_2(\lambda)_j + i\,k(\lambda)_j$$

$$\cos(\beta_j) = \left[1 - \left(\frac{\sin(\alpha_j)}{\tilde{n}_2(\lambda)_j} \right) \right]^{\frac{1}{2}}$$

$$\varrho_j(\alpha,\lambda) = \frac{1}{2} \left(\left| \frac{\tilde{n}_2^j \cos(\alpha_j) - \cos(\beta_j)}{\tilde{n}_2^j \cos(\alpha_j) + \cos(\beta_j)} \right|^2 + \left| \frac{\cos(\alpha_j) - \tilde{n}_2^j \cos(\beta_j)}{\cos(\alpha_j) + \tilde{n}_2^j \cos(\beta_j)} \right|^2 \right)$$

$$\varrho_\odot^j(\alpha) = \frac{\int \varrho(\alpha,\lambda)\, I_\odot(\lambda)\, d\lambda}{\int I_\odot(\lambda)\, d\lambda} \; ; \; \alpha \in [\lambda_{min}; \lambda_{max}]$$

Modelul optic complet pentru repr. $\mathcal{Y}_j$. ↑

$$\varrho_\odot^j = \frac{\bar{\bar{\Phi}}_R^j}{\bar{\bar{\Phi}}_i^j} \Rightarrow \bar{\bar{\Phi}}_R^j = \varrho_\odot^j \cdot \bar{\bar{\Phi}}_i^j$$

$$\bar{\bar{\Phi}}_R = \sum_{j=1}^{N} \bar{\bar{\Phi}}_R^j = \sum_{j=1}^{N} \varrho_\odot^j \bar{\bar{\Phi}}_i^j = \sum_{j=1}^{N} \varrho_\odot^j I_\odot \cos(\alpha_j) S_j \Rightarrow$$

$$\bar{\bar{\Phi}}_i = \sum_{j=1}^{N} I_\odot \cos(\alpha_j)\, S_j$$

$$q_{eff} = \frac{\sum_{j=1}^{M} q_0^j I_0 \cos(\alpha_j) S_j}{\sum_{j=1}^{M} I_0 \cos(\alpha_j) S_j} = \frac{\sum_{1=1}^{M} q_0^j \cos(\alpha_j) S_j}{\sum_{j=1}^{M} \cos(\alpha_j) S_j}$$

↑ coeficientul efectiv global pentru $\{\gamma_j\}_{j=1}^{M}$.

Am considerat incertă dir. de prop. a radiației luminoase. ⇒ Generez puncte de iluminare aleatoare pe o sferă de R=1 în jurul satelitului.

$$\vec{r}(\theta;\phi) = (R\sin(\theta)\cos(\phi); R\sin(\theta)\sin(\phi); R\cos(\theta))^T$$

$$= (\sin(\theta)\cos(\phi); \sin(\theta)\sin(\phi); \cos(\theta))^T$$

$$dS = \left| \frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right| d\theta\, d\phi$$

$$\frac{\partial \vec{r}}{\partial \theta} = (\cos(\theta)\cos(\phi); \cos(\theta)\sin(\phi); -\sin(\theta))^T$$

$$\frac{\partial \vec{r}}{\partial \phi} = (-\sin(\theta)\sin(\phi); \sin(\theta)\cos(\phi); 0)^T$$

$$\Rightarrow u \wedge v \cup [-1,1] \; ; \; \phi \sim \cup [0, 2\pi)$$

$$u = \cos(\theta) \Rightarrow \sin u(\theta) = \sqrt{1-u^2}$$

$$\hat{k} = \left(\sqrt{1-u^2}\cos(\phi) \; ; \sqrt{1-u^2}\sin(\phi) \; ; \; u \right)^T$$

$$\Rightarrow dP \propto |du| \, d\phi$$

$$\Rightarrow \sin(\theta) = -\frac{du}{d\theta} \Rightarrow dP \propto \left| -\frac{du}{d\theta} \right| d\theta d\phi \Rightarrow$$

$$\forall i \; u = \omega(\theta) \Rightarrow du = -\sin(\theta) d\theta \Rightarrow$$

$$+ k \left(\sin(\theta)\cos(\theta)\cos^2(\phi) + \cos(\theta)\sin(\theta)\sin^2(\phi) \right)$$

$$= \hat{i} \sin^2(\theta)\cos(\phi) + \hat{j} \sin^2(\theta)\sin(\phi) + \hat{k} \sin(\theta)\cos(\theta)$$

$$ds = |\sin(\theta)| \, d\theta d\phi$$

$$\text{Jacobian pre pjate} \Rightarrow dP \propto dS \Rightarrow$$

$$\Rightarrow dP \propto |\sin(\theta)| \, d\theta d\phi$$

$$= \hat{i}\left(-\sin^2(\theta)\sin(\phi)\right) - \hat{j}\left(-\sin^2(\theta)\sin(\phi)\right) +$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos(\theta)\cos(\phi) & \cos(\theta)\sin(\phi) & -\sin(\theta) \\ -\sin(\theta)\sin(\phi) & \sin(\theta)\cos(\phi) & 0 \end{vmatrix} =$$

Simulare Monte Carlo

$$\hat{k}_t = \left(\sqrt{1-\mu_t^2}\cos(\phi_t);\ \sqrt{1-\mu_t^2}\sin(\phi_t);\ \mu_t\right)^T$$

$$-\hat{k}_t \cdot \hat{n}_j = \cos(\alpha_{j,t})$$

$$q_t = q_{eff}(\hat{k}_t) = \frac{\sum_{j=1}^{H} q_0^j(\alpha_{j,t})\, s_j \cos(\alpha_{j,t})}{\sum_{j=1}^{H} s_j \cos(\alpha_{j,t})};\ t \in \overline{1,\mathcal{L}}$$

$$\overline{q}_{eff} = \frac{1}{\mathcal{L}} \sum_{t=1}^{\mathcal{L}} q_t \longrightarrow \sigma_q;\ P_5;\ P_{50};\ P_{95}$$